Course outline

About NPTEL

How does an NPTEL online course work?

Week 0

Week 1

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Week 7

- Approximate Differential Privacy
- Exponential Mechanism
- Fairness in Machine Learning
- Week 7- Lecture Materials
- Quiz: Week 7: Assignment 7
- Week 7 Feedback Form: Responsible & Safe AI Systems
- Assignment 7 solution

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Download videos

Week 7: Assignment 7

The due date for submitting this assignment has passed.

Due on 2024-09-

As per our records you have not submitted this assignment.

1) When calculating the sensitivity in ε-Differential Privacy where the values to be derived from the data points is a d-dimension vector, id normalisation technique. (Notations are the same as used in the lecture)

- Manhattan normalisation
- Eucledian normalisation
- Max normalisation
- Min-max normalisation
- Sigmoid normalisation

No, the answer is incorrect.
Score: 0

Accepted Answers:
Manhattan normalisation

2) In (ε, δ)- Differential privacy what does $\delta = 0$ imply? (Notations are the same as used in the lecture)

- The equation $(P(M(x) \epsilon S) \leq e^{\varepsilon} (P(M(x') \epsilon S)$ should hold for some of the subsets S
- The equation $(P(M(x) \epsilon S) \leq e^{\varepsilon} (P(M(x') \epsilon S)$ should hold for most of the subsets S
- The equation $(P(M(x) \epsilon S) \leq e^{\varepsilon} (P(M(x') \epsilon S)$ should hold for all of the subsets S
- The equation $(P(M(x) \epsilon S) \leq e^{\varepsilon} (P(M(x') \epsilon S)$ should hold for none of the subsets S

No, the answer is incorrect.
Score: 0

Accepted Answers:
The equation $(P(M(x) \epsilon S) \leq e^{\varepsilon} (P(M(x') \epsilon S)$ should hold for all of the subsets S

3) How do the utilities vary in the Laplacian mechanism vs the Gaussian mechanism in a higher dimension differential privacy setting?

- As the dimension increases, the Gaussian mechanism requires quadratically more amount of noise than the Laplacian mechanism, de utility
- As the dimension increases, the Gaussian mechanism requires quadratically lesser amount of noise than the Laplacian mechanism, d utility
- As the dimension increases, the Gaussian mechanism requires quadratically lesser amount of noise than the Laplacian mechanism, in utility
- As the dimension increases, the Gaussian mechanism requires quadratically more amount of noise than the Laplacian mechanism, inc utility

No, the answer is incorrect.
Score: 0

Accepted Answers:
As the dimension increases, the Gaussian mechanism requires quadratically lesser amount of noise than the Laplacian mechanism, increas

4) _____ property ensures that a function applied on the privacy-protected data _____ its privacy aspect after applying a function over it.

- i. Post-processing ii. Retains
- i. Post-processing ii. Loses
- i. Composition ii. Retains
- i. Composition ii. Loses

No, the answer is incorrect.
Score: 0

Accepted Answers:
i. Post-processing ii. Retains

5) After using k mechanisms for getting $k(\varepsilon, \delta)$- differentially private data variations for a dataset, the combined leakage that is observed these k mechanisms can be minimized by:

- Using Laplacian Mechanism
- Using Gaussian Mechanism
- Using Uniform Mechanism
- Using Exponential Mechanism

No, the answer is incorrect.
Score: 0

Accepted Answers:
Using Gaussian Mechanism

6) In a buyer-seller problem, given n buyers and n valuations by the buyers, what is the total **revenue** given a price p.

- $p \sum_{i=n}^{n} A \text{ where } A = 1 \text{ if } vi \geq p \text{ and } A = 0 \text{ if } vi \leq p$
- $p \sum_{i=n}^{n} A \text{ where } A = 0 \text{ if } vi \geq p \text{ and } A = 1 \text{ if } vi \leq p$
- pn
- $p(n-1)$
- $p(1/n)$

No, the answer is incorrect.
Score: 0

Accepted Answers:

$p \sum_{i=n}^{n} A \text{ where } A = 1 \text{ if } vi \geq p \text{ and } A = 0 \text{ if } vi \leq p$

7) In the exponential mechanism to calculate the price to maximize the revenue, identify the correct statement in the scenario where 2 un prices result in the same revenue:

- Both prices have an unequal probability of being selected
- Both prices have an equal probability of being selected
- A higher price has a higher probability of being chosen due to normalisation
- A lower price has a higher probability of being chosen due to normalisation

No, the answer is incorrect.
Score: 0

Accepted Answers:
Both prices have an equal probability of being selected

8) In a classification problem, if a data point lies on a hyperplane that perfectly separates the two classes, the probability of the data point belonging to class A is:

- 25%
- 50%
- 75%
- 100%

No, the answer is incorrect.
Score: 0

Accepted Answers:
50%

9) In a vanilla Principle Component Analysis method, the reconstruction loss of a protected group is _____ than the remaining data befo resampling and _____ than the remaining data after resampling.

- Higher, higher
- Higher, lower
- Lower, higher
- Lower, lower

No, the answer is incorrect.
Score: 0

Accepted Answers:
Higher, higher

10) The goal of a Fair PCA is to find a PCA solution U where U=[Ua, Ub] such that reconstruction loss of the two groups A and B where A i protected group is:

- Equal
- Unequal
- The protected group has a lower reconstruction loss
- The protected group has a higher reconstruction loss

No, the answer is incorrect.
Score: 0

Accepted Answers:
Equal

11) In an ideal situation where the models are completely fair, the different parity values are:

- Approach 0
- 1
- Approach 1
- 0

No, the answer is incorrect.
Score: 0

Accepted Answers:
0

12) Match the following:

i. $P(M(x) = 1 \mid x \text{ in } C) - P(M(x) = 1)$	a. Fair Logistic regression

ii. $P(M(x) = 1|y = 1 \text{ and } C) - P(M(x) = 1|y = 1)$ b. Statistical Parity
iii. $P(M(x) = 1|C = 1) - P(M(x) = 1|C = 0)$ c. Equality of Opportunity

- i. - a, ii. - b, iii. - c
- i. - b, ii. - a, iii. - c
- i. - c, ii. - a, iii. - b
- i. - b, ii. - c, iii. - a

No, the answer is incorrect.
Score: 0

Accepted Answers:
i. - b, ii. - c, iii. - a